

# 13 Excretion in human

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## Excretory Products and Organs

The body excretes waste products through several organs:

| Excretory Product                 | Excretory Organ    | Source                                                  |
|-----------------------------------|--------------------|---------------------------------------------------------|
| Carbon Dioxide (CO <sub>2</sub> ) | Lungs              | Waste product of aerobic respiration.                   |
| Urea                              | Kidneys (in urine) | Waste from the breakdown of excess amino acids (toxic). |
| Excess Water & Ions               | Kidneys (in urine) | Substances in excess of the body's requirements.        |

The **Urinary System** is the main route for removing nitrogenous waste and excess water/ions. It consists of the **kidneys, ureters, bladder, and urethra**.

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## The Kidney: Structure and Function

The kidney is the primary organ for filtering the blood and forming urine.

### Kidney Structure

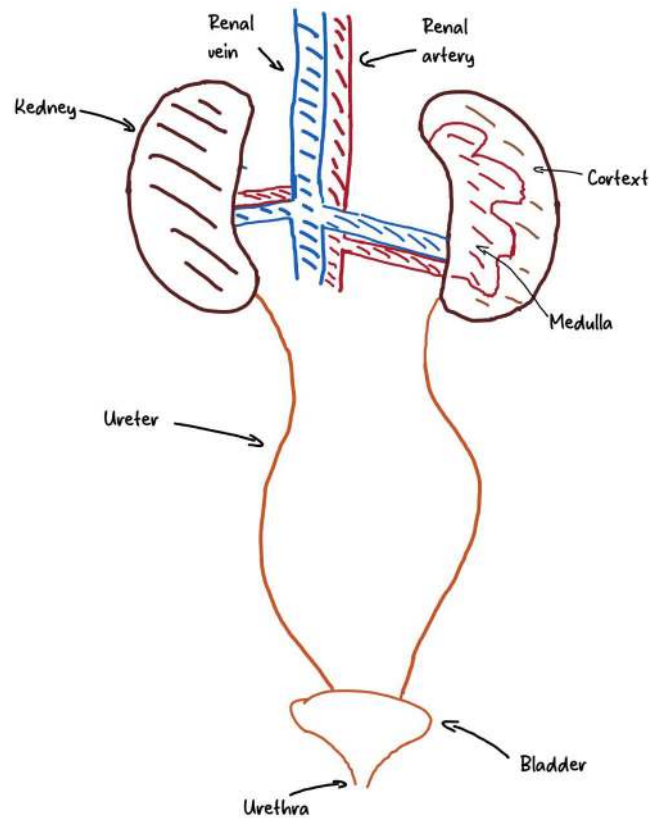
The kidney is composed of two main regions:

- **Cortex:** The outer layer.
- **Medulla:** The inner region.

### Nephron Function (Urine Formation)

The nephron is the functional unit of the kidney, responsible for three main stages of urine formation:

- 1 **Filtration:** The **glomerulus** filters the blood, allowing water, glucose, urea, and ions to pass out of the blood, while blood cells and proteins (large molecules) stay in blood.
- 2 **Reabsorption:** The renal tubules reabsorbs **all of the glucose (active transport)**, some of the ions, and most of the water back into the blood.
- 3 **Urine Formation:** The final fluid is **urine**, which contains urea, excess water, and excess ions.



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## The Role of the Liver

The liver manages the body's amino acid supply and is the site of urea formation.

### Assimilation of Amino Acids

The liver uses amino acids absorbed from the small intestine to synthesise essential proteins, a process known as **assimilation**.

### Urea Formation (Deamination)

When there is an excess of amino acids, the liver converts them into urea:

- 1 **Deamination** is the **removal of the nitrogen-containing part of amino acids to form urea**.
- 2 The amino group is removed, converted into ammonia, and then quickly into the less toxic substance, **urea**.
- 3 The remaining part of the amino acid molecule is converted to carbohydrate and can be respired or stored.